Class 3

Science

Solutions

Olympiad Program

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1. Plants

Worksheet – 1

1. **Correct answer: A**
   Plants are essential because they produce oxygen during photosynthesis and provide food directly or indirectly to all living organisms.
2. **Correct answer: B**
   Roots absorb water and minerals from the soil and supply them to the rest of the plant.
3. **Correct answer: B**
   Mango is a tree because it is tall, has a strong woody stem (trunk), and lives for many years.
4. **Correct answer: B**
   Herbs are small plants with soft, green, and tender stems (e.g., mint, coriander).
5. **Correct answer: B**
   Creepers have weak stems and spread along the ground (e.g., pumpkin, watermelon).
6. **Correct answer: B**
   Leaves prepare food for the plant using sunlight, carbon dioxide, and water. This process is called photosynthesis.
7. **Correct answer: C**
   Money plant is a climber because it needs support to grow upward.
8. **Correct answer: C**
   Rose is a shrub as it is medium-sized and has woody stems branching near the base.
9. **Correct answer: B**
   Trees provide oxygen, food, shelter, and help maintain ecological balance.
10. **Correct answer: C**
    During photosynthesis, plants release oxygen, which is necessary for breathing in humans and animals.

Worksheet – 2

1. **Correct answer: B**
   Roots anchor the plant firmly in the soil, preventing it from being uprooted by wind.
2. **Correct answer: B**
   The shoot system includes all above-ground parts: stem, leaves, flowers, and fruits.
3. **Correct answer: B**
   The stem transports water and minerals from roots and food from leaves to other parts.
4. **Correct answer: C**
   Leaves are called the “kitchen of the plant” because they prepare food through photosynthesis.
5. **Correct answer: A**
   Leaves use water, sunlight, and carbon dioxide to make food.
6. **Correct answer: B**
   Flowers are involved in reproduction and help in seed formation.

7. Correct answer: C
Fruits protect the seeds, which can grow into new plants.

8. Correct answer: C
A mango tree has a tap root system with one main deep root.

9. Correct answer: C
The stem supports the plant and transports water and food.

10. Correct answer: C
Germination is the process by which a seed grows into a new plant.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct answer: B
Tomato plant is a herb because it has a soft, green stem.

2. Correct answer: B
Climbers have weak stems and need support to grow upward.

3. Correct answer: A
Roots absorb water and minerals from the soil.

4. Correct answer: C
Neem is a tree as it is tall with a strong woody trunk.

5. Correct answer: D
Leaves prepare food through photosynthesis.

6. Correct answer: B
Photosynthesis is the process by which plants make their own food using sunlight.

7. Correct answer: A
The stem transports water from roots to leaves.

8. Correct answer: B
The fruit protects the seeds inside it.

9. Correct answer: B
Wheat has a fibrous root system with many thin roots.

10. Correct answer: C
The flower develops into a fruit after fertilization.

11. Correct answer: C
Carrot is a taproot, having one thick main root.

12. Correct answer: B
Flowers are the reproductive parts of a plant.

13. Correct answer: C
The stem does not produce seeds; seeds are formed in flowers.

14. Correct answer: A
Rose is a shrub with woody stems branching near the base.

15. Correct answer: A
Annual plants complete their life cycle in one season.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

16. Correct answer: A, B, C
The shoot system includes leaves, stems, and flowers. Roots belong to the root system.

17. Correct answer: A, B, C
Wind, water, and animals help in dispersal of seeds. The moon has no role.

18. Correct answer: A, C
Leaves perform photosynthesis and transpiration. Water absorption is done by roots, and transport is mainly through the stem.

19. Correct answer: A, C, D
Carrot, radish, and beetroot store food in their roots. Potato stores food in a stem (underground stem/tuber).

20. Correct answer: A, B
Air and water are essential for germination. Sunlight and soil are not strictly necessary for the initial stage.

ASSSERTION AND REASON TYPE

21. Correct answer: C
The assertion is true (roots absorb water and minerals), but the reason is false because photosynthesis occurs in leaves, not roots.

22. Correct answer: C
The assertion is true (seeds need air and water), but the reason is false because sunlight is not essential for germination.

23. Correct answer: A
Both are true, and the reason correctly explains the assertion because trees live for many years, so they are perennial.

24. Correct answer: A
Both are true, and the reason correctly explains the assertion since flowers form fruits and seeds after pollination, enabling reproduction.

25. Correct answer: A
Both are true, and the reason correctly explains the assertion because climbers have weak stems and need support to grow.

MATRIX MATCHING TYPE

26. (a) Root → (Q) Absorbs water
(b) Leaf → (P) Makes food
(c) Flower → (R) Produces seeds
(d) Stem → (S) Supports plant

27. (a) Herb → (S) Wheat
(b) Shrub → (Q) Rose
(c) Climber → (R) Pea
(d) Tree → (P) Neem

COMPREHENSION TYPE

Passage 1

28. Correct answer: A
Seeds need water (along with air and warmth) to begin growing.

29. Correct answer: C
The root (radicle) comes out first to absorb water and anchor the plant.

30. Correct answer: A
Plants grow by getting water, air, and warmth from sunlight, which support germination and further growth.

Passage 2

31. Correct answer: A
Chlorophyll helps in photosynthesis, allowing leaves to make food.

32. Correct answer: B
Leaves take in carbon dioxide from the air.

33. Correct answer: A
Leaves help humans and animals by releasing oxygen into the air during photosynthesis.

PREVIOUS YEAR QUESTIONS

1. Correct answer: C
Rice is obtained from seeds (grains), whereas tea (leaves), coffee (seeds but processed differently), and green gram (seeds/pulses) are grouped differently in this context—rice stands out as the staple cereal grain.

2. Correct answer: C
Hibiscus, Rose, Tulsi, and Jasmine are shrubs (medium-sized plants with woody stems). Others are herbs or climbers.

3. Correct answer: A
Jasmine (shrub) and Pea (climber) form the correct pair.

4. Correct answer: C
Broccoli is a modified flower. Without flowers, a plant cannot produce seeds.

5. Correct answer: A
Coriander is a herb with a soft, green stem.

2. Animals and Birds

Worksheet – 1

1. **Correct answer: B**
   Herbivores depend only on plant materials like grass, leaves, and fruits for their food.
2. **Correct answer: C**
   Carnivores eat the flesh of other animals, and a tiger is a clear example of a meat-eating animal.
3. **Correct answer: C**
   Omnivores eat both plants and animals. Bears can eat fruits, plants, and meat.
4. **Correct answer: B**
   Scavengers like vultures eat dead animals, helping to clean the environment and prevent the spread of diseases.
5. **Correct answer: B**
   Canines are long and sharp teeth that help carnivores tear flesh easily.
6. **Correct answer: B**
   Herbivores use flat molars to grind tough plant materials like grass and leaves.
7. **Correct answer: C**
   A parasite lives inside or on another organism and takes nutrients from it. Tapeworms live inside the host's body.
8. **Correct answer: B**
   Rabbits' teeth keep growing, so they chew hard objects to keep them from becoming too long.
9. **Correct answer: B**
   Crocodiles have an extremely powerful bite, stronger than most other animals.
10. **Correct answer: C**
    Omnivores eat a mixed diet, including both plant and animal food.

Worksheet – 2

1. **Correct answer: B**
   Plants release oxygen during photosynthesis, which is essential for animals to breathe and survive.
2. **Correct answer: A**
   Animals eat fruits and later drop or excrete the seeds in different places, helping plants spread and grow.
3. **Correct answer: C**
   Butterflies transfer pollen from one flower to another while collecting nectar, helping in pollination.
4. **Correct answer: B**
   Decomposers break down dead plants and animals into simpler substances, recycling nutrients back into the soil.
5. **Correct answer: C**
   Without plants, animals would lose their main source of food and oxygen, making survival impossible.
6. **Correct answer: C**
   Producers like grass make their own food using sunlight through photosynthesis.

7. Correct answer: C
Omnivores eat both plant and animal matter, allowing them to survive on a varied diet.

8. Correct answer: B
Scavengers feed on dead animals, helping to clean the environment and prevent disease spread.

9. Correct answer: A
A simple food chain shows the flow of energy from plants (grass) to herbivores (rabbit) to carnivores (eagle).

10. Correct answer: A
The food chain maintains balance in nature by regulating populations and ensuring energy flow between organisms.

Worksheet – 3

1. Correct answer: B
Birds have hollow bones, which make their bodies light and help them fly easily.

2. Correct answer: B
Strong flight feathers on the wings and tail help birds lift off, stay in the air, and control direction.

3. Correct answer: C
Ducks have webbed feet that help them swim efficiently in water.

4. Correct answer: C
Birds use their beaks to pick, tear, and eat food instead of teeth.

5. Correct answer: B
Ostriches are flightless birds; they cannot fly but can run very fast.

6. Correct answer: B
Birds build nests to lay eggs safely and protect their young ones from danger.

7. Correct answer: B
Birds commonly use natural materials like twigs, leaves, and feathers to build their nests.

8. Correct answer: C
The weaver bird is known for making beautifully woven hanging nests.

9. Correct answer: B
Incubation is the process where birds keep their eggs warm until they hatch.

10. Correct answer: C
We should not disturb bird nests, as it can harm the eggs or baby birds.

COMPETITIVE CORNER
SINGLE CORRECT OPTION TYPE

1. Correct answer: C
Omnivores eat both plants and animals, and a bear can survive on a mixed diet.

2. Correct answer: B
Carnivores use long, sharp canines to tear flesh from their prey.

3. Correct answer: A
A correct food chain starts with a producer (grass), followed by a herbivore (deer), and then a carnivore (lion).

4. Correct answer: B
Hollow bones reduce body weight, helping birds to fly easily.

5. Correct answer: B
The gizzard is a strong muscular organ that grinds food in birds since they do not have teeth.

6. Correct answer: C
The Arctic Tern is famous for migrating very long distances between the poles.

7. Correct answer: B
Herbivores have flat teeth that help in grinding plant materials like grass and leaves.

8. Correct answer: B
A bird's tail helps in maintaining balance and changing direction while flying.

9. Correct answer: B
Vultures feed on dead animals, helping to clean the environment.

10. Correct answer: C
Decomposers break down dead plants and animals, returning nutrients to the soil and maintaining the ecosystem.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Correct answers: A, C
Humans and bears are omnivores, meaning they eat both plants and animals.

12. Correct answers: A, D
Hollow bones make birds light, and a streamlined body reduces air resistance, helping in flight.

13. Correct answers: A, B, D
Pigeons, crows, and ostriches build nests, while penguins usually do not build typical nests.

14. Correct answers: A, B, D
Feathers help in flying, keep the bird warm, and in some birds (like ducks) help in swimming.

15. Correct answers: A, B, C
Plants provide oxygen, animals depend on plants for food, and decomposers recycle nutrients—showing interdependence in nature.

ASSERTION AND REASONING TYPE

16. Correct answer: C
Carnivores do have sharp, pointed teeth, but they are used for tearing flesh, not for grinding plant material.

17. Correct answer: A
Birds migrate to avoid harsh weather, and this movement also helps them find better food sources, which explains the reason.

18. Correct answer: A
Eagles have sharp claws and hooked beaks, and these adaptations help them catch and tear their prey effectively.

19. Correct answer: B
Plants release oxygen during photosynthesis, and animals need oxygen to survive, but this is not the direct explanation of the assertion.

20. Correct answer: A
Birds have a four-chambered heart, which supports efficient oxygen supply and helps maintain the high energy needed for flight.

MATRIX MATCH TYPE

21. Correct answer: (a) – R, (b) – P, (c) – Q, (d) – S

22. Correct answer: (a) – R, (b) – P, (c) – S, (d) – Q

COMPREHENSION TYPE

23. Correct answer: C

Birds migrate to farmlands during winter mainly to find food such as insects and pests.

24. Correct answer: C

Birds help maintain ecological balance by controlling insect and pest populations.

25. Correct answer: B

In agricultural fields, insect-eating birds are commonly found as they feed on pests.

26. Correct answer: B

An increase in deer population provides more food for tigers, so their numbers also increase.

27. Correct answer: C

When deer population decreases, tigers may face food shortage, leading to a decline in their population.

28. Correct answer: C

The food chain maintains balance by regulating the population of different organisms in an ecosystem.

PREVIOUS YEAR QUESTIONS

1. Correct answer: D

Down feathers are soft and fluffy, helping to keep the bird's body warm.

2. Correct answer: C

A food chain shows how plants and animals depend on each other for food.

3. Correct answer: B

The unscrambled word is "GOOSE". A goose has holes (nostrils) on either side of its beak.

4. Correct answer: B

The correct word is "EAGLE", which has sharp claws called talons.

5. Correct answer: B

A food chain shows how organisms get food and depend on each other.

3. Food

Worksheet – 1

1. **Correct answer: B**
   Rice is a plant product and a major staple food rich in carbohydrates.
2. **Correct answer: C**
   Cows and buffaloes give milk, which is a nutritious animal product.
3. **Correct answer: B**
   Potato grows underground and is an example of a root/stem vegetable.
4. **Correct answer: D**
   Bees collect nectar from flowers and convert it into honey in the hive.
5. **Correct answer: B**
   Fish come from animals and are a rich source of protein.
6. **Correct answer: B**
   Wheat is obtained from plants and is used to make many food items.
7. **Correct answer: C**
   Carrot is a root vegetable, so we eat its root part.
8. **Correct answer: B**
   Cheese is made from milk, which comes from animals.
9. **Correct answer: C**
   Fish are obtained from water bodies and provide high-quality animal protein.
10. **Correct answer: B**
    Farmers grow crops like pulses, grains, and vegetables in fields.

Worksheet – 2

1. **Correct answer: A**
   Rice is rich in carbohydrates and provides energy to the body.
2. **Correct answer: C**
   Milk contains protein and calcium, which are important for strong muscles and bones.
3. **Correct answer: B**
   Protective foods like fruits and vegetables help us stay healthy and fight diseases.
4. **Correct answer: C**
   Soft drinks do not provide essential nutrients needed for body growth and strength.
5. **Correct answer: B**
   Oranges are rich in vitamins, especially Vitamin C, which keeps skin healthy.
6. **Correct answer: C**
   A balanced diet includes all types of foods—energy-giving, body-building, and protective foods.
7. **Correct answer: C**
   French fries are considered junk food because they are high in fat and low in nutrients.
8. **Correct answer: B**
   Eating too much junk food can lead to weakness, obesity, and health problems.
9. **Correct answer: B**
   Protein-rich foods include pulses, eggs, milk, and meat, which help in body growth.

10. Correct answer: B
Cooking makes food tasty, soft, and easier to digest.

Worksheet – 3

1. Correct answer: B
Cooking makes food tasty, soft, and easier to digest.

2. Correct answer: C
Steaming uses water vapor to cook food without direct contact with water.

3. Correct answer: C
Cucumber is commonly eaten raw and safe to eat without cooking.

4. Correct answer: C
Cooking kills harmful microorganisms and makes food easier to digest.

5. Correct answer: B
Baking is used to prepare cakes and bread.

6. Correct answer: B
Fruits are healthy and rich in essential nutrients.

7. Correct answer: B
Proteins help in building and repairing muscles.

8. Correct answer: D
Water helps in digestion, keeps skin healthy, and removes toxins.

9. Correct answer: B
Gujarat is famous for Dhokla.

10. Correct answer: B
Japan is the origin of Sushi.

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE

1. Correct answer: C
Rice is a plant-based food obtained from crops.

2. Correct answer: B
Carbohydrates give instant energy to the body.

3. Correct answer: C
Steaming helps retain most nutrients as food is not overcooked or exposed to excess oil.

4. Correct answer: C
Consuming too much junk food is unhealthy and can harm the body.

5. Correct answer: B
Dal and roti are common in India, but not typically eaten in Japan.

6. Correct answer: B
Proteins help in growth and repair of body tissues.

7. Correct answer: C
Chips are unhealthy and not a good source of protein.

8. Correct answer: B
Cooking makes food tastier, safer, and easier to digest.

9. Correct answer: C
Fried and processed foods should be eaten in limited amounts.

10. Correct answer: B
Covering food prevents contamination and helps retain moisture.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Correct answers: A, B, D
Rice, bread, and sugar are rich in carbohydrates, which provide energy.

12. Correct answers: A, C, D
Freezing, drying, and adding salt help preserve food by preventing spoilage.

13. Correct answers: A, B
Calcium and Vitamin D are essential for strong and healthy bones.

14. Correct answers: A, B, D
Fruits and vegetables provide vitamins, boost immunity, and help keep the body hydrated.

15. Correct answers: A, C
Roti and dosa are traditional foods from India.

ASSERTION AND REASONING TYPE

16. Correct answer: A
Junk food is high in fats and sugar, which can lead to obesity when eaten in excess.

17. Correct answer: A
Cooking kills germs because heat destroys harmful bacteria present in raw food.

18. Correct answer: A
Water is essential as it helps in digestion and keeps the body hydrated.

19. Correct answer: A
Different cultures have different eating habits because food depends on geography and climate.

MATRIX MATCH TYPE

20. Correct answer:
(a) – S, (b) – Q, (c) – R, (d) – P
Pulses are a protein source, rice provides carbohydrates, cheese is a dairy product, and spinach is rich in iron.

21. Correct answer:
(a) – Q, (b) – S, (c) – R, (d) – P
Milk comes from cows, meat (mutton) comes from animals like sheep, honey is made by bees, and eggs are laid by hens.

COMPREHENSION TYPE

22. Correct answer: B
Fresh vegetables provide essential nutrients and help keep food healthy.

23. Correct answer: B
Drinking water helps in digestion and keeps the body hydrated.

24. Correct answer: C
Junk food is unhealthy and can harm the body if eaten frequently.

25. Correct answer: B
In India, common foods include rice, wheat, pulses, and vegetables.

26. Correct answer: B
Culture influences the types of food people eat in different regions.

27. Correct answer: B
Rice is a staple food in Japan.

PREVIOUS YEAR QUESTIONS

1. Correct answer: D
A balanced diet includes all types of food in the right amount for proper health.

2. Correct answer: B
Using too much refined oil can lead to weight gain due to excess fat intake.

3. Correct answer: A
Puris and samosas are both prepared using the frying method.

4. Correct answer: A
A low-fiber diet is unhealthy because fiber is important for proper digestion.

5. Correct answer: C

4. Human Needs

Worksheet – 1

1. **Correct answer: B**
   Cleanliness helps prevent diseases, promotes health, and creates a pleasant living environment.
2. **Correct answer: B**
   Sunlight and fresh air kill germs and keep the house fresh.
3. **Correct answer: B**
   Using a covered dustbin helps maintain hygiene and prevents the spread of diseases.
4. **Correct answer: B**
   Regular dusting keeps furniture clean and free from dust.
5. **Correct answer: B**
   Separating wet and dry waste helps in proper disposal and recycling.
6. **Correct answer: B**
   Covering food protects it from flies, dust, and contamination.
7. **Correct answer: D**
   Bathrooms and toilets should be cleaned regularly to maintain hygiene.
8. **Correct answer: B**
   Keeping things in their proper places helps maintain cleanliness.
9. **Correct answer: A**
   An organized house makes it easier to find things and saves time.
10. **Correct answer: D**
    All family members should share responsibility for maintaining cleanliness at home.

Worksheet – 2

1. **Correct answer: B**
   We wear clothes to protect our body and stay comfortable in different weather conditions.
2. **Correct answer: C**
   Light cotton clothes help regulate body temperature and keep us comfortable in summer.
3. **Correct answer: B**
   Cotton absorbs sweat and allows air circulation, keeping the body cool.
4. **Correct answer: A**
   Woolen and warm clothes help keep the body warm in winter.
5. **Correct answer: C**
   Raincoats and umbrellas protect us from rain and keep our body dry.
6. **Correct answer: C**
   Waterproof boots are suitable for the rainy season as they keep feet dry.
7. **Correct answer: B**
   Doctors wear white coats to protect their clothes from germs and stains.
8. **Correct answer: C**
   Students wear school uniforms at school.
9. **Correct answer: A**
   Washing and storing clothes properly helps maintain cleanliness and hygiene.

10. Correct answer: A
Clean clothes improve appearance and prevent the spread of germs.

Worksheet – 3

1. Correct answer: B
An occupation is the work people do to earn a living.

2. Correct answer: B
Farming is a primary occupation because it involves getting raw materials directly from nature.

3. Correct answer: C
Doctors diagnose illnesses and provide treatment to patients.

4. Correct answer: C
Factory workers make useful products in industries.

5. Correct answer: A
Carpenters make furniture using wood.

6. Correct answer: B
Teaching is a tertiary occupation as it provides services.

7. Correct answer: A
A postman delivers letters and parcels.

8. Correct answer: B
Police officers maintain law and order and ensure public safety.

9. Correct answer: A
People choose occupations based on their interests and skills.

10. Correct answer: B
Every occupation plays an important role in society and helps maintain balance.

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE

1. Correct answer: B
Keeping a house clean helps prevent diseases and promotes healthy living.

2. Correct answer: B
Cotton is best for summer as it keeps the body cool and absorbs sweat.

3. Correct answer: C
Farming is a primary occupation because it involves using natural resources.

4. Correct answer: B
Garbage should be put in a covered dustbin to maintain hygiene.

5. Correct answer: B
Woolen clothes keep us warm and protect us from cold weather.

6. Correct answer: C
Secondary occupations involve using raw materials to make finished goods.

7. Correct answer: C
Leaving food uncovered attracts flies and spreads germs.

8. Correct answer: C
Carpenters make and repair wooden furniture.

9. Correct answer: C
Raincoats and synthetic clothes protect us from rain.

10. Correct answer: B
Teachers provide education to students.

11. Correct answer: B
Washing clothes removes dirt and germs, keeping them clean.

12. Correct answer: B
Covering food protects it from flies and dust.

13. Correct answer: C
Shopkeepers sell goods to customers.

14. Correct answer: C
All houses should be cleaned regularly to maintain hygiene.

15. Correct answer: C
Cotton clothes absorb sweat and keep the body cool in hot weather

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. Correct answers: A, B, D
Keeping a house clean prevents diseases, keeps pests away, and improves hygiene.

17. Correct answers: A, C
Proper disposal of garbage keeps surroundings clean and prevents bad odor.

18. Correct answers: A, C, D
Fishing, farming, and mining are primary occupations as they involve natural resources.

19. Correct answers: A, B, D
Teachers, engineers, and doctors provide services to people.

20. Correct answers: A, B, D
Washing, proper storage, and ironing help maintain cleanliness and durability of clothes.

ASSERTION AND REASONING TYPE

21. Correct answer: A
Cotton clothes are worn in summer because they absorb sweat and keep the body cool.

22. Correct answer: A
A clean house promotes good health, and a dirty house attracts germs and pests that cause diseases.

23. Correct answer: C
Woolen clothes are not suitable for summer; they keep the body warm, not cool.

24. Correct answer: A
Teachers are important because they help students gain knowledge and develop skills.

25. Correct answer: C
Farmers belong to primary occupations, but they do not manufacture goods—they produce raw materials.

MATRIX MATCH TYPE

26. Correct answer:
(a) – P, (b) – Q, (c) – R, (d) – S
Farmers grow crops, carpenters make wooden furniture, doctors treat sick people, and shopkeepers sell goods.

27. Correct answer:
(a) – P, (b) – Q, (c) – R, (d) – S
Cotton clothes are worn in summer, woolen clothes in winter, raincoats in the rainy season, and silk clothes on special occasions.

COMPREHENSION TYPE

28. Correct answer: B
John's friend was impressed because the house was clean and well-maintained.

29. Correct answer: B
They keep the house clean by sweeping, dusting, arranging things, and disposing of garbage properly.

30. Correct answer: B
Cleanliness prevents germs and diseases, helping people stay healthy.

31. Correct answer: B
Different occupations help society function smoothly by fulfilling various needs.

32. Correct answer: A
Farmer, doctor, and teacher are examples of important occupations.

33. Correct answer: C
Without farmers, there would be no food production.

PREVIOUS YEAR QUESTIONS

1. Correct answer: C
A green grocer sells fruits and vegetables.

2. Correct answer: D
Mud is used for kutcha houses, not pucca houses which use bricks, cement, and iron.

3. Correct answer: A
"CANAAVR" unscrambles to "CARAVAN," which is a temporary house.

4. Correct answer: D
"RELPMUB" unscrambles to "PLUMBER," who fixes leaking taps.

5. Correct answer: D
Pihu is most likely in the bedroom as she is still waking up.

5. Transport, Communication, and Safety Rule

Worksheet – 1

1. **Answer: C) Air Transport**
   Air transport is the fastest mode of transport.
2. **Answer: B) Rocket**
   Rockets are used for traveling in space.
3. **Answer: D) Train**
   Trains run on tracks and are used for long-distance travel.
4. **Answer: C) Seat belt**
   Seat belts protect passengers and improve safety in cars.
5. **Answer: C) Metro**
   Metro trains are used for underground travel in big cities**.**
6. **Answer: B) Stop**
   A red traffic light means stop.
7. **Answer: C) Train**
   A train is a means of land transport.
8. **Answer: D) Bicycle**
   Bicycles do not use fuel and do not cause pollution.
9. **Answer: B) Train**
   A train is a land transport vehicle suitable for city travel.
10. **Answer: C) Zebra crossing**
    Pedestrians should use zebra crossings to cross roads safely.

Worksheet – 2

1. **Answer: A) To express thoughts and ideas**
   Communication is used to share thoughts, ideas, and information.
2. **Answer: C) Smiling at a friend**
   Smiling conveys a message without words, so it is non-verbal communication.
3. **Answer: B) Mobile phone**
   A mobile phone is used for speaking and listening, which is verbal communication.
4. **Answer: B) It can be saved and read later**
   Written messages can be stored and referred to in the future.
5. **Answer: A) Pigeon post**
   Pigeon post is an old method of communication, not a modern one**.**
6. **Answer: D) The person in figure b can move without remaining connected to cable**
   Wireless communication allows movement without cable connections.
7. **Answer: A) Share personal details with strangers**
   Sharing personal information online with strangers is unsafe.
8. **Answer: A) Verbal communication**
   TV news mainly communicates through spoken words.
9. **Answer: D) Password**
   A password is for security, not a means of communication**.**
10. **Answer: C) Being respectful and avoiding fake news**
    Safe internet communication involves respectful behavior and verifying information.

Worksheet – 3

1. **Answer: B) School is at North-East to the post office**
   The school is located above and to the right of the post office, which is North-East.
2. **Answer: C) Left**
   Left is a relative direction, not a main cardinal direction.
3. **Answer: B) To tell the directions**
   A compass rose shows directions such as North, South, East, and West.
4. **Answer: C) Roads, highways, and travel routes**
   A road map mainly shows roads and routes used for travel.
5. **Answer: B) To explain symbols used on the map**
   The legend key helps users understand map symbols.
6. **Answer: C) Compass**
   A compass is used to find directions.
7. **Answer: B) East**
   When facing North, East is on the right side.
8. **Answer: B) Physical map**
   Physical maps show natural features like mountains, rivers, and lakes.
9. **Answer: B) Different places and landmarks**
   Map symbols represent places, landmarks, and features.
10. **Answer: B) Google Maps**
    Google Maps is a commonly used digital navigation tool.

Worksheet – 4

1. **Answer: B) Electrical switches**
   Wet hands can cause electric shock when touching electrical switches.
2. **Answer: C) Out of reach of children**
   Prevents accidental poisoning and injuries.
3. **Answer: C) Hold the railing**
   Helps prevent slips and falls on stairs.
4. **Answer: B) Use a zebra crossing**
   It is the safest place for pedestrians to cross the road.
5. **Answer: A) Stop**
   A red traffic light means stop and wait.
6. **Answer: C) A helmet**
   Protects the head from injuries while riding.
7. **Answer: A) Apply antiseptic and bandage**
   Cleans the wound and helps prevent infection.
8. **Answer: B) Apply ice pack**
   Reduces pain and swelling immediately after a bruise.
9. **Answer: B) Ice pack**
   Cold treatment helps reduce swelling.
10. **Answer: B) To prevent accidents and injuries**
    Safety rules protect people from harm.

Practice Corner

Single Correct Option Type

1. **Answer: A) Compass rose**
   It helps identify directions and locate places on a map.
2. **Answer: B) Pour cold water**
   Cold water cools the burn and reduces damage**.**

3. Answer: C) Talking on a phone call
Verbal communication uses spoken words.

4. Answer: B) Train
A train is a public transport system used by many people.

5. Answer: D) Mirror
A mirror is not a component of a map.

6. Answer: B) Scale
A scale converts map distance into actual distance.

7. Answer: C) Find an alternate route
A "No Entry" sign means you must not enter that road.

8. Answer: B) Satellite
Satellites transmit signals for live television broadcasts.

9. Answer: B) Clean the wound and stop bleeding
This helps prevent infection and control blood loss.

10. Answer: A) Aeroplane
It is an air transport vehicle, while the others are land transport vehicles.

ONE OR MORE THAN CORRECT OPTION TYPE

11. Answer: A, B, C
Television, radio, and newspapers are mass communication media because they reach large audiences. A personal letter is meant for individual communication.

12. Answer: A, B, D
Keeping electrical appliances away from water, locking doors when alone, and storing medicines out of children's reach improve home safety. Leaving sharp objects on the floor is unsafe**.**

13. Answer: A, C
Email and social media are modern digital communication methods. Pigeons and telegram services are older forms of communication.

14. Answer: A, B, D
First aid involves staying calm, providing immediate help, and seeking medical assistance when necessary. Delaying treatment is not a proper first-aid practice.

ASSERTION AND REASON TYPE

15. Answer: A
Digital maps are more convenient because they provide real-time navigation and traffic updates, making travel easier and more efficient.

16. Answer: C
First aid helps reduce the severity of injuries, but it can be given by trained people and is not restricted only to doctors.

Matching Type Questions

17. Answer: A–P, B–Q, C–R, D–S
Airplane → international travel; Newspaper → mass communication; Traffic Light → road safety regulation; First Aid → immediate care for injuries.

18. Answer: A–P, B–Q, C–R, D–S
Political maps show countries and states; Physical maps show natural features; Road maps show roads and highways; Weather maps show climatic conditions.

COMPREHENSION TYPE

Passage-I

19. Answer: B) The signal turned red
Ravi stopped because he saw the traffic signal turn red.

20. Answer: C) Safe crossing
A zebra crossing is meant to help pedestrians cross the road safely.

21. Answer: B) Guiding children
The passage states that the policeman was guiding school children on the footpath.

Passage-II

22. Answer: B) To find route
Meera used a map to find the shortest route for her trip.

23. Answer: A) Show way
Landmarks help travelers identify locations and directions during navigation.

24. Answer: B) Avoid traffic
Meera checked distances and landmarks to avoid traffic congestion while traveling.

Previous year questions

1. Answer: D) Internet
Sending an e-mail requires an internet connection.

2. Answer: D) All of these
The pictures show unsafe behavior such as creating disturbance in class, playing near a road, and unsafe conduct in a vehicle.

3. Answer: C) 6
The Postal Index Number (PIN) used in India consists of 6 digits.

4. Answer: C) Boat
Boats are commonly used for traveling short distances in water bodies.

5. Answer: D) Cinema and Post card
Cinema is a means of mass communication that reaches many people simultaneously, while a **Post card** is a means of personal communication intended for a specific individual.

6. Human Body

Worksheet – 1

1. Correct answer – B

Digestion begins the moment you take a bite. Your teeth start mechanical digestion, and enzymes in your saliva start chemical digestion.

2. Correct answer – A

The stomach secretes hydrochloric acid (HCl) and pepsin, creating a high-acid environment (pH of about 1.5 to 3.5) specifically to break down protein chains.

3. Correct answer – A

By chewing, teeth break large chunks of food into tiny particles. This increases the total surface area, allowing digestive enzymes to work much faster.

4. Correct answer – C

Your incisors are the sharp, chisel-shaped teeth at the front of your mouth, designed specifically for biting into and cutting food.

5. Correct answer – C

While the stomach starts the process, the small intestine is the "heavy lifter." About 90% of digestion and nearly all nutrient absorption happens here.

6. Correct answer – A

Saliva contains an enzyme called amylase, which begins breaking down complex carbohydrates (starches) into simpler sugars.

7. Correct answer – B

The stomach acts as a temporary holding tank. It can expand to store a meal while it slowly turns it into a liquid called "chyme" before passing it to the intestine.

8. Correct answer – B

Canines are the pointed, "fange-like" teeth. Their sharp, conical shape is evolved for gripping and tearing tougher foods like meat.

9. Correct answer – B

By the time food reaches the large intestine, most nutrients are gone. Its main job is to reclaim water and electrolytes to keep you hydrated and solidify waste.

10. Correct answer – C

The process is a "one-way street." It begins with ingestion (eating) and ends with the excretion of indigestible materials (feces) through the rectum and anus.

Worksheet – 2

1. Correct answer – B

This system is responsible for gas exchange—bringing in oxygen (O_2) and removing carbon dioxide (CO_2).

2. Correct answer – B

These are the external entry points where air is filtered, warmed, and moistened before heading to the lungs.

3. Correct answer – B

Its job is to support cellular respiration by providing the necessary oxygen.

4. Correct answer – B

This dome-shaped muscle sits below the lungs. When it contracts (moves down), it creates a vacuum that pulls air in.

5. Correct answer – B

While we exhale some leftover oxygen and nitrogen, carbon dioxide is the primary waste product removed during exhalation.

6. Correct answer – A
Think of the circulatory system as the body's delivery and waste-pickup service.

7. Correct answer – C
The heart is a muscular pump that generates the pressure needed to keep blood moving through miles of vessels.

8. Correct answer – B
Arteries carry blood Away from the heart. Most (except the pulmonary artery) carry oxygen-rich blood.

9. Correct answer – C
Capillaries are microscopic and have walls only one cell thick, allowing nutrients and oxygen to pass easily into tissues.

10. Correct answer – B
Veins carry blood back to the heart under low pressure. They have one-way valves to prevent blood from flowing backward.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct answer – B
Teeth perform mechanical digestion, physically breaking food down to increase surface area for enzymes.

2. Correct answer – C
This is where the majority of chemical breakdown occurs, aided by bile and pancreatic juices.

3. Correct answer – B
Thanks to tiny finger-like projections called **villi**, the small intestine is perfectly designed to absorb nutrients into the blood.

4. Correct answer – B
Specifically, "Salivary Amylase," which breaks down starches into simpler sugars while you chew.

5. Correct answer – C
These are tiny air sacs at the end of the bronchioles where the actual exchange of gases with the blood takes place.

6. Correct answer – B
The primary goal is gas exchange; circulation (A) and transport (D) are the jobs of the circulatory system

7. Correct answer – B
By changing the volume of the chest cavity, the diaphragm forces air into and out of the lungs.

8. Correct answer – B
Arteries have thick, elastic walls to handle the high pressure of blood being pumped directly from the heart.

9. Correct answer – B
The heart's cardiac muscle works 24/7 to ensure blood reaches every cell in your body.

10. Correct answer –B
Red blood cells contain **hemoglobin**, a protein that binds to oxygen and gives blood its red color.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, B
Living things grow and reproduce. Dissolving and melting are physical changes usually associated with non-living matter.

12. Correct answer – B, C
Eyes are the organs for vision; the **Tongue** is the organ for taste.

13. Correct answer – A, D
The nervous system coordinates all body actions and specifically manages reflex actions (fast, automatic responses).

14. Correct answer – C, D
The male reproductive system produces **Sperms** (cells) and **Testosterone** (the primary male sex hormone).

15. Correct answer – A, C
The system takes in **Oxygen** and removes **Carbon Dioxide**.

ASSERTION AND REASONING TYPE

16. Correct answer – A
Villi significantly increase the surface area, which is exactly why the small intestine is so efficient at absorption.

17. Correct answer – D
Assertion is False: The heart pumps *deoxygenated* blood to the lungs to get oxygen. It pumps oxygen-rich blood to the *body*.

18. Correct answer – A
Haemoglobin is the specific molecule that "grabs" oxygen to transport it, making the reason a perfect explanation.

19. Correct answer – A
Exhalation is the primary mechanism the respiratory system uses to dump CO_2 waste.

20. Correct answer – B
Both are true, but regulating water balance/blood pressure is a separate function from "filtering toxins," so R doesn't "explain" A.

MATCH THE FOLLOWING TYPE

21. Correct answer – (a)-(R), (b)-(S), (c)-(P), (d)-(Q)

22. Correct answer – (a)-(S), (b)-(P), (c)-(Q), (d)-(R)

COMPREHENSHION TYPE

Passage1

23. Correct answer – B) Transports oxygen and nutrients through the blood

24. Correct answer – B) By providing oxygen to the blood and removing carbon dioxide

25. Correct answer – C) To break down food and absorb nutrients

Passage2

26. Correct answer – B) To filter toxins, waste, and excess substances from the blood

27. Correct answer – B) By maintaining fluid balance in the body

28. Correct answer – B) Kidneys, ureters, bladder, and urethra

Previous Year Questions

1. **Correct answer – A**
   The doorbell is a sound stimulus, so she used her sense of **Hearing**.
2. **Correct answer – D**
   The brain controls actions, handles memory, and receives sensory messages.
3. **Correct answer – B**
   The **Skeletal System** provides the frame for movement and protects organs (like the skull protecting the brain).
4. **Correct answer – C**
   Incorrect statement: Digestion *starts* in the mouth and continues in the stomach, but **complete** digestion occurs in the small intestine.
5. **Correct answer – A**
   Physical exercise like **Playing football** strengthens the musculoskeletal system.

7. Earth and Universe

Worksheet – 1

1. Answer: A
The Sun is a star. The others are not stars.

2. Answer: B) Mars
Mars is called the Red Planet because of its reddish surface.

3. Answer: A) Saturn
Saturn has the most prominent ring system in the Solar System.

4. Answer: B) Jupiter
Jupiter is the largest planet in the Solar System**.**

5. Answer: B) 8
There are 8 planets in the Solar System.

6. Answer: C) Jupiter
The Great Red Spot is a giant storm on Jupiter.

7. Answer: D) Mercury
Mercury is the closest planet to the Sun.

8. Answer: A) 1 and 5
1 (Earth) and **5 (Saturn)** are planets, which directly orbit the Sun. The others are either distant stars, parts of Earth's atmosphere (clouds), or a moon orbiting a planet.

9. Answer: B) Constellation
A pattern of stars in the sky is called a constellation**.**

10. Answer: A) 3 → 1 → 2
A star is smaller than a constellation, and a galaxy is the largest.

Worksheet – 2

1. Answer: B) Earth's rotation on its axis
Earth rotates on its axis, causing half of it facing the sun (day) and other half not (night).

2. Answer: B) 24 hours
Earth takes about 24 hours to complete one rotation on its own axis.

3. Answer: B) Orbit
The path Earth follows around the Sun is called its orbit.

4. Answer: C) Earth's tilt on its axis and its revolution around the Sun
Seasons occur due to Earth's tilt and its movement around the Sun.

5. Answer: A) New Moon
About 14–15 days after a Full Moon, the Moon reaches the New Moon phase**.**

6. Answer: A) New Moon
During a New Moon, the Moon is not visible from Earth.

7. Answer: B) Only a small part of the Moon is visible
In the waxing crescent phase, a small illuminated portion of the Moon is visible and increasing.

8. Answer: A) 1 and 2
Statements 1 and 2 are correct; statement 3 is incorrect because waxing crescent means the illuminated part is increasing.

9. Answer: C) Communication and weather study
Satellites are used for communication and weather monitoring.

10. Answer: B) Using a rocket
Rockets launch artificial satellites into space.

Worksheet – 3

1. Answer: B) Sputnik 1
Sputnik 1, launched by the Soviet Union in 1957, was the first artificial satellite.

2. Answer: B) Yuri Gagarin
Yuri Gagarin became the first human in space in 1961.

3. Answer: A) Apollo 11
Apollo 11 was the first mission to land humans on the Moon in 1969.

4. Answer: B) Mars
Curiosity and Perseverance are rovers that explore Mars.

5. Answer: C) Galileo Galilei
Galileo improved the telescope and discovered four major moons of Jupiter.

6. Answer: B) Gravity
The Sun's gravity keeps planets moving in their orbits.

7. Answer: A) ISS
The International Space Station (ISS) is where astronauts live and work in space.

8. Answer: B) Copernicus
Copernicus proposed that Earth and other planets revolve around the Sun.

9. Answer: B) Johannes Kepler
Kepler discovered that planets move in elliptical orbits.

10. Answer: B) Traveling beyond Earth's atmosphere
Space travel means traveling outside Earth's atmosphere into space.

Competitive Corner

Single Correct Option Type

1. Answer: D) Gibbous
The Moon marked 'X' is where More than half of the Moon can be seen.

2. Answer: C) Cassiopeia
The constellation Cassiopeia is easily recognized by its distinctive "W" shape in the night sky.

3. Answer: B) Orbit
The fixed path followed by a planet around the Sun is called its orbit.

4. Answer: B) Moon
The Moon is the large rocky natural satellite that revolves around Earth.

5. Answer: C) Core
The Sun's core is the central region where nuclear fusion takes place, producing energy.

6. Answer: D) Saturn
Saturn currently has the highest known number of moons among all planets in the Solar System.

7. Answer: B) It reflects sunlight
The Moon does not produce its own light; it appears bright because it reflects sunlight.

8. Answer: B) When one celestial object blocks another
An eclipse occurs when one celestial body moves into the shadow of or blocks another celestial body.

9. Answer: A) Venus
Venus is the hottest planet because its thick carbon dioxide atmosphere traps heat through a strong greenhouse effect.

10. Answer: B) R

In the diagram, the largest outer region represents the Solar System. The object inside it but not inside the planet–moon group is the Sun, represented by R.

ONE OR MORE THAN CORRECT OPTION TYPE

11. Answer: A, C

The Moon orbits Earth and has virtually no atmosphere. It is smaller than Earth and does not have weather like Earth.

12. Answer: A, C, D

The Sun is a star, the largest object in the Solar System, and its gravity keeps the planets in orbit. It is mainly made of hydrogen and helium, not water.

13. Answer: B, C

Jupiter and Saturn are gas giants. Mars and Venus are rocky terrestrial planets.

14. Answer: A, B, D

Mars, Earth, and Jupiter have moons. Venus has no natural moons.

ASSERTION AND REASON TYPE

15. Answer: C) Assertion is true and Reason is false.

Earth's tilt causes seasons, but Earth's axis remains tilted, not upright.

16. Answer: B) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

The Moon revolves around Earth, and its gravity causes tides, but tides do not explain why the Moon revolves around Earth.

Matching Type Questions

17. Answer: A–Q, B–S, C–R, D–P

The Sun is the center of the Solar System, the Moon is Earth's natural satellite, Earth is our home planet, and the Asteroid Belt lies between Mars and Jupiter.

18. Answer: A–Q, B–P, C–P, D–R

Mars has 2 moons, Jupiter has over 90 moons, Earth has 1 moon. The Saturn has over 280 moons which means more than 90 moons.

COMPREHENSION TYPE

Passage-I

19. Answer: B) Causes tides

The Moon's gravitational pull causes tides in Earth's oceans.

20. Answer: C) Meteorite impacts

The Moon's craters were formed by meteorite impacts.

21. Answer: C) 27.3 days

The Moon takes about 27.3 days to complete one orbit around Earth.

Passage-II

22. Answer: B) It gives light and heat

The Sun provides light and heat that support life on Earth.

23. Answer: B) Earth's movement around the Sun

The passage states that Earth's movement around the Sun causes the seasons.

Previous year questions

1. **Answer: C) Orion**
   Orion is called the Hunter constellation because its star pattern resembles a hunter.
2. **Answer: B) R has many huge craters created by space rocks.**
   In the figure, R represents the Moon, which has numerous craters formed by impacts of meteoroids and other space rocks.
3. **Answer: A) New Moon**
   About 14–15 days after a Full Moon, the Moon reaches the New Moon phase.
4. **Answer: D) Both B and C**
   The figure shows a Lunar Eclipse, and statements B and C are correct. Therefore, the statement "Both B and C" as incorrect is the correct choice.
5. **Answer: A) New Moon**
   The given phase is a Waning Crescent. The phase that immediately follows it is the New Moon.

8. Matter and Minerals

Worksheet – 1

1. **Answer: c) Tiny particles**
   Matter is anything that has mass and occupies space. All matter is made up of tiny particles called atoms and molecules.
2. **Answer:** b) Mass
   Mass is the measure of the amount of matter in an object and tells us how heavy it is.
3. **Answer:** b) Volume
   Volume measures how much space an object occupies.
4. **Answer:** c) They have a fixed shape and volume
   Solids have a definite shape and volume because their particles are closely packed and vibrate in place.
5. **Answer:** b) Liquid
   Liquids do not have a fixed shape but take the shape of the container they are in. Their volume, however, remains constant.
6. **Answer:** c) Temperature
   Changes in temperature can cause matter to change from one state to another, e.g., solid → liquid → gas.
7. **Answer:** c) Water droplets forming on a cold glass
   Condensation occurs when water vapor in the air cools down and changes back into liquid water.
8. **Answer:** c) Because of evaporation
   Steam is formed when the liquid water in the hot tea evaporates into water vapor.
9. **Answer:** c) A balloon bursting
   Bursting a balloon is a physical change but not a change of state. Changes of state involve solid, liquid, and gas transitions.
10. **Answer:** C) Beaker R: A cup of fruit salad with grapes and apples
    A heterogeneous mixture has visibly different substances that are not uniformly distributed. In fruit salad, you can see the different fruits. The others are either solutions (homogeneous) or mixtures of gases that are uniform.

Worksheet – 2

1. **Answer: c) Cotton**
   Natural materials are obtained from plants, animals, or minerals. Cotton comes from cotton plants, so it is natural. Plastic, glass, and steel are man-made.
2. **Answer: b) Glass**
   Transparent materials allow light to pass through. Glass is commonly used for windows because it is clear and lets light in
3. **Answer: c) Paper**
   Man-made materials are created by humans from natural materials. Paper is made from wood pulp, so it is man-made. Wool and leather are natural, clay is natural too.
4. **Answer: b) Rubber**
   Tires are made from rubber because it is elastic, strong, and can absorb shocks.
5. **Answer: d) Plastic**
   Waterproof materials do not allow water to pass through. Plastic is waterproof, whereas paper, wood, and cotton absorb water.

6. Answer: b) Transparency
Transparency is the property of a material that allows light to pass through it, like glass or clear plastic.

7. Answer: c) Wood
Paper is made from the pulp of wood, which is processed and pressed into sheets.

8. Answer: b) Rubber
Flexible materials can bend easily without breaking. Rubber is highly flexible, unlike stone, glass, or iron.

9. Answer: b) Cotton
Cotton is soft, breathable, and comfortable to wear, making it ideal for clothing.

10. Answer: C) A Glass pane
Glass is man-made, transparent, and brittle. When struck hard, it shatters into many pieces. Steel and rubber do not shatter like glass, and wood is not transparent.

Worksheet – 3

1. Answer: c) Centimeter
Small lengths, like a pencil, are measured in centimeters (cm). Kilogram measures weight, liter measures volume, and hour measures time.

2. Answer: c) Clock
A clock or watch measures time. Ruler measures length, weighing scale measures weight, and measuring cup measures volume.

3. Answer: b) Kilogram
Kilogram (kg) is used to measure heavier objects like watermelons. Gram is used for very light objects, liter for liquids, and centimeter for length.

4. Answer: b) Liter
Liter (L) is the standard unit to measure the volume of liquids. Kilogram is for mass, meter is for length, gram is for mass.

5. Answer: b) 100
1 meter = 100 centimeters.

6. Answer: c) A stopwatch
A stopwatch measures precise time in seconds and fractions of seconds. A wall clock shows general time, measuring tape measures distance, and thermometer measures temperature.

7. Answer: c) Kilometer
Kilometers (km) are used to measure long distances. Millimeter and centimeter are too small, gram measures weight.

8. Answer: b) 60
1 minute = 60 seconds.

9. Answer: c) Weighing scale
A weighing scale measures mass. Ruler measures length, measuring tape measures length, stopwatch measures time.

10. Answer: d) Meter
Taller objects, like trees, are measured in meters. Centimeter is used for smaller objects, kilogram measures weight, milliliter measures volume.

Competitive Corner

Single answer correct type:

1. Answer: c) Melting
Melting is when a solid absorbs heat and changes into a liquid. Freezing is the

opposite, condensation changes gas to liquid, and evaporation changes liquid to gas.

2. Answer: c) Iron
Metals like iron conduct heat well, whereas plastic, wood, and rubber are poor conductors (insulators).

3. Answer: b) Meter
Meter (m) is the standard unit of length in the metric system.

4. Answer: c) Water
Water is liquid at room temperature. Ice is solid, steam is gas, and oxygen is a gas.

5. Answer: b) Measuring cup
Measuring cups or graduated cylinders are used to measure liquid volume. Ruler measures length, stopwatch measures time, weighing scale measures mass.

6. Answer: c) It evaporates into steam
Water boils at 100°C and changes from liquid to gas (steam).

7. Answer: d) Rubber
Insulators resist heat flow. Rubber, plastic, and glass are poor conductors, but rubber is particularly effective as a heat insulator.

8. Answer: a) Kilogram
Heavier objects like potatoes are measured in kilograms. Gram is used for very light objects.

9. Answer: c) 1000
1 liter = 1000 milliliters.

10. Answer: c) Evaporation
Evaporation is the process where liquid water changes into gas (steam) due to heat.

11. Answer: b) Ductility and Thermal conductivity
Ductility is the ability to be stretched into thin wires. Thermal conductivity is the ability to transfer heat.

12. Answer: B) The components are distributed uniformly throughout the sample.
A homogeneous mixture has uniform composition throughout, meaning the components are evenly distributed.

13. Answer: B) Evaporation → Condensation
In distillation, the liquid first evaporates into vapor, and then the vapor condenses back into liquid in another container.

Multiple correct answer:

14. Answer: a) Glass, b) Plastic sheet, d) Air
Transparent materials allow light to pass through so that objects can be seen clearly. Glass, clear plastic sheets, and air are transparent. Wood is opaque and does not let light pass through.

15. Answer: a) Ruler, b) Weighing scale, c) Stopwatch
Measuring instruments are tools used to find the value of a quantity.
Ruler → measures length
Weighing scale → measures mass/weight
Stopwatch → measures time
A pencil is not a measuring instrument.

16. Answer: a) Plastic bottle, c) Wooden block
Objects float if they are less dense than water. Plastic and wood are less dense, so they float. Iron and stone are denser than water, so they sink.

17. Answer: a) Plastic, c) Glass, d) Paper

Explanation: Man-made materials are created by humans using natural resources.

- Plastic → made from petroleum
- Glass → made from sand and other chemicals
- Paper → made from wood pulp

Cotton is a natural material obtained from cotton plants.

Assertion and reason type:

18. Answer: C) Assertion is true and Reason is false.

The **assertion is true** because gases can be compressed easily.

The **reason is false** because gas molecules are **far apart**, not tightly packed. The large spaces between molecules allow gases to be compressed.

19. Answer: A) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

The **assertion is true**: Measuring cups are used to measure liquid volume.

The **reason is true** and correctly explains the assertion: Measuring cups are marked with units like liters (L) and milliliters (mL), which allows accurate measurement of liquids.

Match the following:

20. **(a) Liquid →(R) Water:** Water at room temperature has a definite volume but no fixed shape, allowing it to flow and take the shape of its container. This is the defining characteristic of a liquid.

(b) Gas → (Q) Water vapor: Water vapor consists of molecules that are far apart and move rapidly. It has neither a fixed shape nor a fixed volume, filling whatever space it is in.

(c) Solid → (P) Ice: When water freezes, its molecules lock into a rigid structure. Ice maintains its own shape regardless of the container, making it a solid.

(d) Changing states → (S) Evaporation: This is the specific physical process where a substance changes from a liquid state to a gaseous state, usually due to an increase in temperature.

21. (a) Length → (P) Ruler: A ruler (or measuring tape) is used to determine the distance between two points. In the lab, we usually measure this in centimeters (cm) or millimeters (mm).

(b) Mass → (Q) Weighing scale: Mass is the amount of "stuff" or matter in an object. We use a scale to measure this, typically in grams (g) or kilograms (kg).

(c) Time → (R) Clock: To observe how fast a reaction happens or how long an experiment takes, we use a clock or stopwatch to measure duration in seconds, minutes, or hours.

(d) Volume → (S) Measuring cylinder: Volume is the amount of space an object (usually a liquid) occupies. A graduated measuring cylinder allows us to read the level of a liquid precisely, usually in milliliters (mL).

Comprehension type:

Passage-1

22. Answer: B) A solid changes into a liquid

Ice is a solid. When it absorbs heat, it melts and becomes liquid water. This is a change of state from solid → liquid.

23. Answer: C) Evaporation

Evaporation is the process where liquid water changes into water vapor (gas) due to heat. Freezing and solidification are liquid → solid, and condensation is gas → liquid.

24. Answer: D) It has no fixed shape or volume and can spread to fill any container

Gas particles move freely and occupy the space available. They do not have a fixed shape or volume.

Passage-2

25. Answer: C) Milliliters (mL)

Explanation: Milliliters (mL) are used to measure small volumes of liquid. Kilometers measure distance, meters measure length, and milligrams measure mass.

26. Answer: B) A weighing scale

Explanation: A weighing scale measures mass or weight. A ruler measures length, a measuring spoon measures small volumes, and a clock measures time.

27. Answer: D) A clock or watch

Explanation: Clocks and watches are instruments for measuring time. Thermometers measure temperature, and measuring jugs measure liquid volume.

Previous Year Questions

1. Answer: b) Generally the unit for measuring mass of light objects is gram.

2. Answer: d) All of these

3. Correct Answer: b) X

4. Correct Answer: d) metre

5. Correct Answer: d) Carbon dioxide, Brick, Juice

9. Light, Sound and Force

Worksheet – 1

1. **Answer: B) Sun**
   The Sun is the main natural source of light on Earth.
2. **Answer: B) Torch**
   A torch is a man-made source of light.
3. **Answer: C) It bounces back**
   Light reflects (bounces back) from a mirror.
4. **Answer: B) Glass window**
   A glass window is transparent and allows light to pass through completely.
5. **Answer: B) Shadow**
   A shadow forms when an object blocks light.
6. **Answer: C) Firefly**
   A firefly is a living organism that produces its own light.
7. **Answer: C) Clear plastic**
   Clear plastic is transparent and lets light pass through.
8. **Answer: A) Reflection**
   Reflection is the bouncing back of light from a surface.
9. **Answer: D) Wind**
   Wind is not related to the behavior of light.
10. **Answer: C) Object blocked more light**
    Moving the torch closer makes the object block more light, creating a larger shadow.

Worksheet – 2

1. **Answer: B) Vibrations**
   Sound is produced when objects vibrate.
2. **Answer: C) Birds chirping**
   Birds chirping is a natural source of sound**.**
3. **Answer: C) Ears**
   We use our ears to hear sounds**.**
4. **Answer: C) Drum**
   A drum produces a loud sound when it is struck.
5. **Answer: B) High-pitched**
   A whistle produces a high-pitched sound.
6. **Answer: C) Solid**
   Sound travels fastest through solids because their particles are closely packed.
7. **Answer: B) Sound becomes louder**
   Increasing vibrations increases the loudness of sound.
8. **Answer: A) Both are sounds**
   Music and noise are both forms of sound.
9. **Answer: C) Air, water and solids**
   Sound can travel through gases, liquids, and solids.
10. **Answer: B) No air in space**
    Sound cannot travel without a medium, and there is no air on the Moon.

Worksheet – 3

1. **Answer: B) A push or pull**
   Force is a push or pull that can change an object's motion.
2. **Answer: D) Tugging a rope**
   Tugging a rope is an example of a pull.
3. **Answer: C) Gravity**
   Gravity pulls objects toward the Earth.
4. **Answer: B) A car stopping when brakes are applied**
   Friction between the brakes and wheels slows the car.
5. **Answer: A) Muscular force**
   Riding a bicycle requires force from our muscles**.**
6. **Answer: C) Magnetic force**
   Magnets attract iron objects using magnetic force.
7. **Answer: C) Gravity pulling an apple down**
   Gravity acts without physical contact.
8. **Answer: B) Slows down and stops**
   Friction causes the toy car to slow down and stop.
9. **Answer: B) Friction acts**
   Friction and air resistance slow the cycle when pedaling stops.
10. **Answer: A) A boy pushing a door**
    The boy is using his muscles to push the door. This is a clear example of **muscular force**.

Competitive Corner

Single Correct Option Type

1. **Answer: B) Translucent objects**
   Translucent objects allow some light to pass through but not completely.
2. **Answer: D**
   At noon, the Sun is overhead, so the shadow is the shortest.
3. **Answer: B) Vibrates**
   Sound is produced when an object vibrates.
4. **Answer: A) Pushing a door**
   A force is a push or pull; pushing a door is an example of force.
5. **Answer: D) Muscular**
   We use muscular force to pick up and move objects**.**
6. **Answer: B) Gravitational force**
   Gravity pulls objects towards the Earth.
7. **Answer: A) A shadow is formed**
   A shadow forms when an object blocks light.
8. **Answer: B) Whisper**
   A whisper is a soft sound.
9. **Answer: B) Lubricants like oil**
   Lubricants reduce friction between surfaces.
10. **Answer: B) Light travels along a straight line**
    The candle is visible through the straight pipe but not through the curved pipe, showing that light travels in straight lines.

ONE OR MORE THAN CORRECT OPTION TYPE

11. Answer: A, B, D
Torch, Sun, and Firefly produce their own light, so they are luminous objects. The Moon only reflects sunlight.

12. Answer: A, B, C, D
All these materials reflect light. Mirrors and shiny metals reflect strongly, while glass and paper also reflect some light.

13. Answer: A, B
The size of a shadow depends on the distance of the object from the light source and the size/shape (type) of the object. Color and brightness do not affect the shadow's size.

ASSERTION AND REASON TYPE

14. Answer: D) Assertion is false and Reason is true.
Sound travels fastest in solids, not in water. Water molecules are closer together than air molecules, so the reason is true.

15. Answer: C) Assertion is true and Reason is false.
Force can make an object move or stop. Force can be felt through its effects, so the reason is false.

Matching Type Questions

16. Answer: (a)–P, (b)–Q, (c)–R, (d)–S
Sun is luminous, glass is transparent, brakes work using friction, and a guitar produces sound.

17. Answer: (a)–P, (b)–Q, (c)–R, (d)–S
Mirrors reflect light, wood is opaque, a falling apple shows gravity, and drums produce sound through vibrations.

COMPREHENSION TYPE

Passage-I

18. Answer: A) Change in Sun's position
The Sun's changing position changes the angle of light, causing the shadow size to change.

19. Answer: B) Very short
When the Sun is overhead, the shadow becomes shortest.

20. Answer: C) Same.
The sun is so incredibly far away from the Earth that its light rays are essentially parallel by the time they reach us.

Passage-II

21. Answer: A) Vibrations
The vibrating guitar strings produce sound.

22. Answer: A) Changes pitch
Tightening or loosening a string changes its vibration frequency and therefore its pitch.

23. Answer: A) Drum
A drum produces sound when its membrane vibrates.

Previous Year Questions

1. **Answer: C) Frictional force is less on the wet floor**
   Water reduces friction between the feet and the floor, making it easier to slip.
2. **Answer: D) Frictional force**
   Friction acts opposite to the direction of motion and opposes movement between surfaces in contact.
3. **Answer: D) Both B and C**
   At 11:00 A.M., the shadow should be shorter and in a different direction. Both the direction and shape/length of the shadow shown are incorrect.
4. **Answer: A**
   The shadow is a solid square because the opaque metal cube is the last object in the light's path, completely blocking the light before it reaches the screen.
5. **Answer: C) Butter paper**
   Butter paper is translucent and allows some light to pass through, whereas the others are opaque.

10. Our Environment

Worksheet - 1

1. Correct Answer: B

Oxygen is essential for human survival because it is used in respiration to release energy from food. Without oxygen, body cells cannot function properly. Carbon dioxide is a waste gas, not used for breathing.

2. Correct Answer: B

Air is invisible under normal conditions, which is why we cannot see it. However, it occupies space, has weight, and exerts pressure. These are all important properties of air.

3. Correct Answer: A

Ice is the solid form of water formed when water freezes at 0°C. Water is the liquid state, and water vapour is the gaseous state. Aqua is just another name for water, not a state.

4. Correct Answer: B

Statements 1, 2, and 4 are correct because air occupies space, humidity measures water vapour, and a rain gauge measures rainfall. Statement 3 is incorrect because nitrogen is actually the most abundant gas in the atmosphere, not the least.

5. Correct Answer: A

Weather refers to the day-to-day condition of the atmosphere, including temperature, humidity, and rainfall. It can change frequently. Climate, on the other hand, is the long-term pattern.

6. Correct Answer: B

Seasons change because the Earth is tilted on its axis and revolves around the Sun. This causes different parts of the Earth to receive varying amounts of sunlight during the year. Rotation alone does not cause seasons.

7. Correct Answer: C

Summer is the hottest season in most parts of the world because the Sun's rays fall more directly on the Earth. This increases the temperature. Days are also longer during this season.

8. Correct Answer: C

The monsoon season is characterized by heavy rainfall, strong winds, and cloudy skies. It is important for agriculture in many regions. It is not associated with snow or dry weather.

9. Correct Answer: A

People wear woolen clothes in winter because they keep the body warm by trapping heat. Wool acts as an insulator. Cotton and light clothes are more suitable for summer.

10. Correct Answer: B

A tornado is an extreme weather event with very strong rotating winds that can cause destruction. It is much more severe than a mild breeze or a cloudy day. Extreme weather events are usually dangerous.

Worksheet - 2

1. Correct Answer: A

Granite is a natural rock formed from cooled magma, making it an igneous rock. Gold, diamond, and copper are minerals or elements, not rocks. Rocks are made up of one or more minerals.

2. **Correct Answer: B**

Marble is a metamorphic rock formed when limestone undergoes heat and pressure. This process changes its structure and properties. Sandstone and limestone are sedimentary, while granite is igneous.

3. Correct Answer: B

Mining is the process of removing valuable minerals and rocks from beneath the Earth's surface. It is done using tools and machinery. It is not related to oceans or weather study.

4. Correct Answer: A

Igneous rocks are formed when molten lava or magma cools and solidifies. This can happen either below or above the Earth's surface. The cooling process gives these rocks their structure.

5. **Correct Answer: B**

Statements 1, 2, and 4 are correct because soil forms from weathering, gains nutrients from decayed matter, and is improved by organisms. Statement 3 is incorrect because parent rock is the bottom layer, not the topmost fertile layer.

6. Correct Answer: B

Soil is a mixture of minerals, organic matter (like humus), air, and water. These components support plant growth. It is not made up of only one or two materials.

7. Correct Answer: C

The parent rock layer is the bottommost layer of soil and contains large pieces of rock. It is less weathered compared to upper layers. Topsoil and subsoil have smaller particles.

8. Correct Answer: B

Earthworms improve soil fertility by breaking down organic matter and mixing it with soil. They also create tunnels that help air and water circulate. This makes the soil healthier for plants.

9. Correct Answer: C

Loamy soil is a balanced mixture of sand, clay, and humus. It holds enough water while allowing proper air circulation. This makes it ideal for farming.

10. Correct Answer: B

Clayey soil has very fine particles that hold water tightly. It retains more water than sandy or loamy soil. However, it may have poor drainage.

Worksheet – 3

1. Correct Answer: B

Pollution refers to the contamination of the environment by harmful substances. It affects air, water, and land, making them unsafe for living beings. Clean and healthy surroundings are the opposite of pollution.

2. Correct Answer: B

Vehicles and factories release harmful gases like carbon monoxide and smoke into the air. These pollutants degrade air quality and harm health. Planting trees actually helps reduce pollution.

3. Correct Answer: C

Throwing plastic and garbage into rivers contaminates water bodies. It harms aquatic life and makes water unsafe for use. This type of pollution directly affects water quality.

4. Correct Answer: B

Noise pollution is caused by loud and disturbing sounds. High-volume loudspeakers can harm hearing and cause stress. Natural sounds like birds chirping are not considered pollution.

5. Correct Answer: B

Using bicycles reduces the emission of harmful gases from vehicles. This helps in lowering air pollution levels. Cutting trees and burning garbage increase pollution instead.

6. Correct Answer: B

Excessive honking and loudspeakers create unwanted sound in the environment. This leads to noise pollution, which can affect human health. It causes irritation and hearing problems.

7. Correct Answer: C

Improper disposal of garbage leads to land pollution. Plastic and waste accumulate and harm soil quality. It also affects plants, animals, and humans.

8. Correct Answer: B

Underground mining involves digging deep tunnels to reach minerals below the Earth's surface. It is used when minerals are located far beneath the ground. Open-pit mining is used for surface deposits.

9. Correct Answer: C

Air pollution can cause breathing problems such as asthma. Polluted air irritates the lungs and respiratory system. Other diseases listed are usually caused by contaminated food or water.

10. Correct Answer: C

Planting trees improves air quality and reduces pollution. Proper waste disposal prevents land and water pollution. These practices help maintain a clean and healthy environment.

Competitive Corner

Single Option Correct Type

1. Correct Answer: B

Nitrogen is the most abundant gas in Earth's atmosphere, making up about 78% of the air. Oxygen is the second most abundant. Carbon dioxide and hydrogen are present in much smaller amounts.

2. Correct Answer: B

Evaporation is the process in which liquid water changes into water vapour due to heat. Condensation is the opposite process (vapour to liquid). Precipitation refers to rain, and sublimation is solid directly to gas.

3. Correct Answer: A

Rocks are made of minerals, so P = made of minerals. Both rocks and minerals are naturally occurring substances, so Q fits that region. Minerals have a definite composition, which correctly matches R.

4. Correct Answer: C

Sandstone is a sedimentary rock formed from layers of sand particles compressed over time. Granite and basalt are igneous rocks, while marble is a metamorphic rock. Sedimentary rocks are usually formed by deposition.

5. Correct Answer: B

Humus is the dark, organic matter present in the top layer of soil. It is formed from the decay of plants and animals. Humus makes the soil fertile and suitable for plant growth.

6. Correct Answer: C

Loamy soil is a mixture of sand, clay, and humus, making it ideal for crops. It retains enough water while allowing proper air circulation. This balance supports healthy plant growth.

7. Correct Answer: B

Fish breathe oxygen that is dissolved in water using their gills. Even though the glass looks full of water, it still contains dissolved air. This allows fish to survive and breathe normally.

8. Correct Answer: B

Statement (i) is correct as minerals are natural substances that form rocks. Statement (iii) is correct because minerals are widely used in making jewellery, tools, and glass. Statement (iv) is also correct since quartz is used in glass and watches, while (ii) is incorrect because minerals have a definite composition.

9. Correct Answer: C

Recycling and composting reduce the amount of waste on land. They help reuse materials and convert organic waste into useful manure. This reduces pollution and protects the environment.

10. Correct Answer: B

Using public transport and bicycles reduces fuel consumption and air pollution. It decreases harmful emissions from vehicles. Cutting trees and burning fuels increase pollution instead.

One or More than One Correct Option Type:

11. Correct Answer: A, B, C

Oxygen, carbon dioxide, and nitrogen are major gases present in the atmosphere.

12. Correct Answer: A, B, C

Weather is mainly affected by temperature, air pressure, and humidity. These factors determine daily atmospheric conditions like rain, wind, and heat. Soil type does not directly influence weather.

13. Correct Answer: A, B, C

Excessive construction, mining, and dumping waste or chemicals all damage the land and cause pollution. These activities degrade soil quality and harm living organisms. Planting more trees actually helps reduce land pollution.

Assertion and Reason Type

14. Correct Answer: A

Both the assertion and reason are true, and the reason correctly explains the assertion. Air is important for living beings because it contains oxygen, which is essential for breathing and survival.

15. Correct Answer: A

Both the assertion and reason are true, and the reason correctly explains the assertion. Water pollution reduces oxygen levels in water, which makes it difficult for aquatic organisms to survive.

Matrix Matching Type

16. Correct Answer: a-R, b-S, c-P, d-Q

Igneous rocks include granite, formed from cooled magma. Metamorphic rocks include marble, formed under heat and pressure. Sedimentary rocks include sandstone, formed from sediments, and humus is part of soil layers.

17. Correct Answer: a–R, b–P, c–S, d–Q

Tornadoes are fast-moving spirals of wind. Cyclones bring strong winds and heavy rain. Floods happen when too much rain causes rivers to overflow, while droughts occur due to no rain for a long time.

Comprehension Type

Passage – I

18. Correct Answer: B

The change of seasons is mainly caused by the tilt of the Earth's axis as it revolves around the Sun. This tilt leads to varying sunlight in different regions. Rotation only causes day and night, not seasons.

19. Correct Answer: C

When a hemisphere is tilted towards the Sun, it receives more direct sunlight. This results in warmer temperatures and longer days. Therefore, that hemisphere experiences summer.

20. Correct Answer: D

When one hemisphere experiences summer, the opposite hemisphere is tilted away from the Sun. It receives less sunlight and has shorter days. This leads to winter conditions.

Passage – II

21. Correct Answer: B

Water is necessary for all living beings to survive. It is used in vital processes like digestion, circulation, and temperature regulation. It is not limited to just one activity like farming or cleaning.

22. Correct Answer: B

Shortage of clean water is mainly caused by pollution and excessive use of water resources. Contaminated water becomes unsafe for use. Overuse reduces the availability of fresh water.

23. Correct Answer: B

Cleaning requires the use of water for washing and maintaining hygiene. Activities like playing games, flying airplanes, or reading books do not require water. Hence, cleaning is the correct choice.

Previous Year Questions

1. Correct Answer: A

Chalk is a type of soft sedimentary rock, which makes it easy to write with on blackboards. Its softness allows it to leave marks without damaging the surface. It is not hard like granite or made of silica like quartz.

2. Correct Answer: D

Water vapour from the heated water rises and comes in contact with the cold metallic plate (cooled by ice). This causes the vapour to cool and change into water droplets. This process is called condensation, and the droplets form on the underside of the plate.

3. **Correct Answer: B**

The correct word formed is DEW, which are water droplets seen on leaves during cold mornings. Dew forms due to condensation of water vapour. The letters WDE rearrange to form DEW.

4. Correct Answer: D

Weather is affected by factors like the Sun, rain, and clouds. These influence temperature, humidity, and precipitation. Rocks do not directly affect weather conditions.

5. Correct Answer: C

Drying of clothes happens due to evaporation and does not require additional water. Activities like bathing, cooking, and washing clothes all need water. Hence, drying does not require water.